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# NETGUARD AI: DASHBOARD RINGAN BERBASIS MACHINE LEARNING UNTUK DETEKSI ANOMALI JARINGAN DAN PEMANTAUAN RISIKO

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**Abstrak**--Deteksi anomali jaringan penting bagi institusi pendidikan yang mengelola laboratorium komputer, server lokal, dan gerbang internet dengan anggaran monitoring terbatas. Artikel ini menyajikan NetGuard AI, yaitu sistem machine learning berbiaya rendah untuk mendeteksi trafik jaringan anomali dan memvisualisasikan risiko operasional melalui dashboard ringan. Penelitian ini menggunakan subset representatif dari trafik DDoS CICIDS2017 Friday Afternoon. Tahap preprocessing menormalkan nama kolom, mengubah label menjadi kelas biner, menangani nilai hilang dan infinity, mempertahankan fitur numerik, menghapus duplikasi, serta mengeksport data terproses untuk eksperimen yang dapat direproduksi. Logistic Regression, Decision Tree, dan Random Forest dibandingkan menggunakan accuracy, precision, recall, F1-score, dan confusion matrix. Model terbaik dipilih menggunakan F1-score sebagai kriteria utama, recall sebagai kriteria kedua, dan accuracy sebagai kriteria ketiga. Eksperimen menggunakan 19.935 record terproses dengan train-test split 80:20 dan random\_state=42. Decision Tree memperoleh accuracy 0,9992, precision 0,9995, recall 0,9990, dan F1-score 0,9992. Sistem mengeksport metrik, perbandingan model, confusion matrix, hasil prediksi, dan ringkasan penelitian. Kebaruan penelitian ini adalah integrasi deteksi anomali berbasis machine learning, klasifikasi risiko otomatis, rekomendasi tindakan, dan dashboard web ramah mahasiswa yang dapat direproduksi pada laptop biasa tanpa API berbayar.

**Kata kunci**--deteksi anomali jaringan, CICIDS2017, machine learning, intrusion detection system, dashboard risiko

## I. Pendahuluan

Artificial Intelligence telah mengubah cara trafik jaringan dianalisis. Pada jaringan komputer tradisional, administrator memeriksa log, peringatan firewall, dan grafik bandwidth secara manual. Pendekatan tersebut berguna, tetapi dapat menjadi lambat ketika jumlah paket, flow, atau pengguna meningkat. Institusi pendidikan menghadapi persoalan tambahan karena membutuhkan kemampuan monitoring, namun tidak selalu memiliki infrastruktur keamanan tingkat enterprise.

Proyek NetGuard AI dirancang untuk konteks tersebut. Sistem ini menghubungkan pengetahuan praktis jaringan dengan supervised machine learning sehingga mahasiswa dapat memahami bagaimana fitur network flow diubah menjadi hasil prediksi. Proyek ini juga dirancang sebagai artefak penelitian yang mencakup dokumentasi dataset, script preprocessing, script training, output evaluasi, visualisasi dashboard, dan draft artikel yang dapat dikembangkan untuk penelitian selanjutnya.

Rumusan masalah utama adalah bagaimana pipeline machine learning berbiaya rendah dapat mendeteksi trafik anomali dan menyajikan hasilnya dalam bentuk yang berguna bagi administrator jaringan pemula. Rumusan ini sengaja bersifat praktis. Proyek tidak hanya melaporkan performa numerik, tetapi juga mengubah anomaly ratio menjadi risiko Low, Medium, atau High serta memberikan rekomendasi tindakan operasional.

## II. Tinjauan Pustaka

Research on network intrusion detection commonly uses public datasets such as CICIDS2017, CICIDS2018, UNSW-NB15, and NSL-KDD [1], [2], [12]. Recent work shows that classical machine learning remains competitive for structured flow features, especially tree-based classifiers [3], [4], [10]. Decision Tree and Random Forest are frequently used because they can capture non-linear patterns and are easier to interpret than many deep learning architectures.

Several studies also warn that benchmark datasets must be used carefully [1], [2]. Dataset leakage, duplicate records, improper train-test separation, and unbalanced class distribution can produce inflated results [1], [2]. Therefore, this project explicitly documents preprocessing and uses a reproducible train-test split. The project also states its limitation: the current experiment focuses on a CICIDS2017 DDoS subset, so future work should include more attack classes and cross-dataset validation.

The literature gap identified in this study is not merely model accuracy [3]-[10]. Many IDS papers present high metrics but provide limited operational packaging for beginners [4], [7], [8]. NetGuard AI positions itself as a reproducible educational monitoring prototype that links model evaluation, dashboard interpretation, risk score, recommendation text, and artifact collection.

## III. Research Gap dan Novelty

The research gap is derived from the literature mapping: first, many studies focus on model comparison but do not provide a lightweight dashboard for education; second, several works use advanced architectures that are less suitable for low-budget laptops; third, reported metrics are often separated from practical response recommendations; fourth, reproducibility artifacts are not always organized for student-level replication.

The main novelty of NetGuard AI is the integration of machine learning anomaly detection with a lightweight educational network monitoring dashboard. The secondary novelty is automated risk classification and actionable recommendations for beginner network administrators. This novelty is realistic because it can be implemented using Python, Scikit-Learn, Flask, Bootstrap, and Chart.js without paid services.

The proposed Research Method is RM1: develop a reproducible supervised machine learning pipeline for binary network anomaly detection using CICIDS2017. RM2: integrate the best model output

into a dashboard and report export system that presents risk level and recommended action.

#### IV. Metode yang Diusulkan

Alur kerja yang diusulkan terdiri dari akuisisi dataset, preprocessing, training model, evaluasi model, ekspor prediksi, visualisasi dashboard, dan pengemasan artefak penelitian. Raw traffic data is stored separately from processed data. The preprocessing stage detects the target column named label, converts the label into 0 for normal traffic and 1 for anomaly traffic, converts numeric features, replaces infinity values, fills missing values, and removes duplicate records.

The training stage compares Logistic Regression, Decision Tree, and Random Forest. Logistic Regression acts as a linear baseline. Decision Tree provides an interpretable non-linear baseline. Random Forest acts as an ensemble baseline that usually improves stability. The best model is saved as a pickle file, and every metric is exported into JSON and CSV so the dashboard and report can read the same experiment result.

The dashboard stage reads reports/metrics.json, reports/model\_comparison.csv, reports/prediction\_result.csv, and reports/research\_summary.json. The risk score is computed from anomaly ratio. Low risk is defined as anomaly ratio below 10 percent, Medium risk from 10 percent to below 30 percent, and High risk at 30 percent or above.

#### V. Skenario Eksperimen

The experiment uses a representative subset of CICIDS2017 Friday-WorkingHours-Afternoon-DDoS [12]. The processed data contains 19,935 records: 9,935 normal records and 10,000 anomaly records. The train-test split is 80:20 with random\_state=42.

All models are evaluated using accuracy, precision, recall, F1-score, and confusion matrix. Accuracy measures overall correctness. Precision measures how many predicted anomalies are truly anomalies. Recall measures how many true anomalies are detected. F1-score balances precision and recall, which is important in anomaly detection because false negatives can be operationally dangerous.

The experiment is intentionally low-budget. It can be executed on a normal Windows laptop using Python and open-source libraries. No paid API, GPU, commercial SIEM, or database server is required for the core experiment.

**TABLE I. LITERATURE MAPPING SUMMARY**

| No | Year | Dataset                                 | Method                                        | Observed Limitation                                            |
|----|------|-----------------------------------------|-----------------------------------------------|----------------------------------------------------------------|
| 1  | 2021 | CICIDS2017                              | Dataset troubleshooting and validation        | Focuses on dataset analysis, not an educational dashboard.     |
| 2  | 2024 | Network intrusion datasets              | Machine learning comparison                   | Operational dashboard integration is limited.                  |
| 3  | 2024 | CICIDS2017                              | Critical dataset review                       | Does not propose a beginner-ready monitoring prototype.        |
| 4  | 2024 | CICIDS2017 / related intrusion datasets | Feature engineering and classifier evaluation | Deployment and reproducible student artefacts are not central. |
| 5  | 2024 | CICIDS2017 / cybersecurity dataset      | Random Forest and permutation importance      | Does not translate interpretation into dashboard actions.      |
| 6  | 2024 | Network intrusion datasets              | Adversarial robustness                        | Not focused on low-budget educational                          |

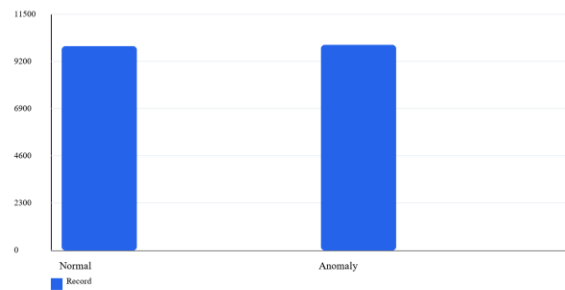
|   |      |                        | evaluation                                  | deployment.                                                  |
|---|------|------------------------|---------------------------------------------|--------------------------------------------------------------|
| 7 | 2025 | Public IDS datasets    | Machine learning / deep learning comparison | Dashboard and novice administrator guidance remain limited.  |
| 8 | 2025 | IDS benchmark datasets | Machine learning comparison                 | Does not emphasize reproducible local-laptop implementation. |

The mapping indicates that the selected studies already cover dataset quality, machine learning comparison, feature importance, ensemble learning, deep learning, and robustness [1]-[10]. However, they do not fully answer the educational deployment need targeted in this study. NetGuard AI therefore narrows its contribution to reproducible implementation, dashboard integration, and operational interpretation rather than claiming a new algorithmic breakthrough.

**TABLE II. EXPERIMENTAL SETUP**

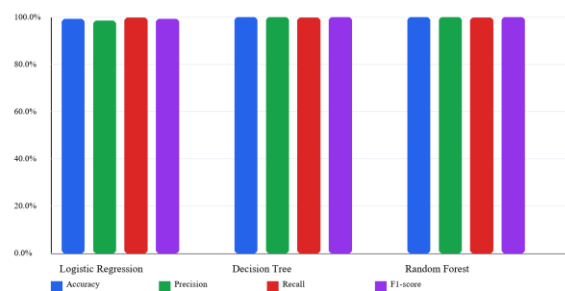
| Parameter         | Value                            |
|-------------------|----------------------------------|
| Dataset           | CICIDS2017 Friday Afternoon DDoS |
| Processed records | 19,935                           |
| Normal records    | 9,935                            |
| Anomaly records   | 10,000                           |
| Split             | 80:20, random_state=42           |
| Selection rule    | F1-score, Recall, Accuracy       |

**Distribusi Dataset CICIDS2017 Subset**



**Fig. 1. Class distribution of CICIDS2017 representative subset.**

**Perbandingan Model NetGuard AI**



**Fig. 2. Performance comparison across Logistic Regression, Decision Tree, and Random Forest.**

#### VI. Hasil dan Pembahasan

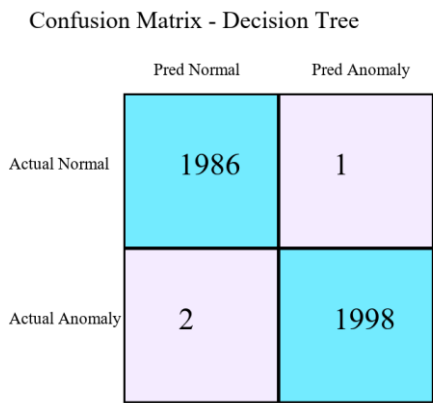
**TABLE III. MODEL COMPARISON**

| Model               | Acc.   | Prec.  | Recall | F1     |
|---------------------|--------|--------|--------|--------|
| Logistic Regression | 0.9920 | 0.9857 | 0.9985 | 0.9921 |
| Decision Tree       | 0.9992 | 0.9995 | 0.9990 | 0.9992 |
| Random Forest       | 0.9992 | 1.0000 | 0.9985 | 0.9992 |

The best model is Decision Tree. It achieved accuracy 0.9992, precision 0.9995, recall 0.9990, and F1-score 0.9992. Decision Tree slightly outperformed Random Forest based on the F1-score selection rule, while Random Forest reached perfect precision in this

split. Logistic Regression also performed strongly, which indicates that the selected CICIDS2017 DDoS flow features are highly separable after preprocessing.

The confusion matrix shows very low misclassification. However, the high score must not be interpreted as universal deployment readiness. The result is specific to the selected CICIDS2017 DDoS subset. In real networks, traffic can change over time, attacks can be mixed, and benign traffic can be more diverse. Therefore, future validation should include cross-validation, multiple CICIDS2017 attack days, UNSW-NB15, CICIDS2018, and local campus traffic where ethical permission is available.



Interpretasi: kesalahan klasifikasi sangat rendah pada data uji subset CICIDS2017.

Fig. 3. Confusion matrix of the selected best model.



Catatan: skor ini berbasis korelasi absolut pada subset tes; gunakan feature importance model untuk analisis lanjutan.

Fig. 4. Feature association analysis for model interpretation.

Risk Score Based on Anomaly Ratio

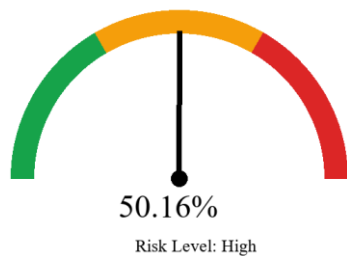


Fig. 5. Risk score visualization based on anomaly ratio.

VII. Analisis Detail Pemilihan Baseline

Logistic Regression is used as the first baseline because it represents a simple linear classifier [13]. If Logistic Regression performs well, the dataset may contain features that separate normal and attack traffic clearly in a linear or near-linear decision space. In this experiment, Logistic Regression achieved a high F1-score, which indicates that many CICIDS2017 DDoS patterns are already distinguishable after preprocessing. However, its lower precision compared with tree-based methods shows that linear boundaries may still misclassify some normal flows as anomalies.

Decision Tree is selected as the second baseline and practical proposed model because it can model non-linear decision rules without requiring heavy computation [10], [13]. It is also easier to explain in an educational context. A student can understand that the tree splits traffic based on feature thresholds. This makes Decision Tree suitable for a low-budget project that must be interpretable during defense. The model obtained the highest F1-score in the current split, making it the selected best model according to the predefined rule.

Random Forest is selected as the third model because it represents an ensemble of many decision trees [5], [10], [13]. It usually improves generalization and reduces variance compared with a single tree. In this experiment, Random Forest achieved perfect precision but slightly lower recall than Decision Tree. This means Random Forest avoided false positives very well, but it missed a small number of anomaly records. Since anomaly detection prioritizes balanced detection and missed attacks are risky, the F1-score rule still selected Decision Tree.

VIII. Interpretasi Detail Metrik Evaluasi

Accuracy is useful as a general measurement, but it can be misleading in imbalanced datasets. If a dataset contains far more normal records than anomaly records, a model can obtain high accuracy by mostly predicting the majority class. Therefore, accuracy is not used as the only selection criterion. In NetGuard AI, accuracy is placed as the tertiary criterion after F1-score and recall.

Precision is important because false alarms can burden network administrators. A low-precision model may label normal traffic as anomaly too often, causing alert fatigue. In a school or campus environment, alert fatigue is dangerous because staff may start ignoring warnings. Random Forest obtained the highest precision in this experiment, which makes it attractive for scenarios where false alarms must be minimized.

Recall is important because a false negative means anomaly traffic is not detected. In network security, missed attacks can be more harmful than false alarms. This is why recall becomes the secondary selection criterion. Decision Tree achieved recall 0.9990, while Logistic Regression achieved 0.9985 and Random Forest achieved 0.9985. The difference is small, but the selection rule is intentionally strict.

F1-score is used as the primary criterion because it balances precision and recall. This makes it suitable for binary anomaly detection where both false positive and false negative errors matter. The Decision Tree F1-score is slightly higher than Random Forest, so it is selected as the best model even though the numerical difference is very small.

## IX. Analisis Error

The confusion matrix of the best model contains a very small number of errors. False positive records indicate normal traffic predicted as anomaly. These errors can cause unnecessary inspection, but they are still manageable when the number is low. False negative records indicate anomaly traffic predicted as normal. This error type is more critical because it can allow suspicious traffic to pass without attention.

For research interpretation, the low number of errors suggests that CICIDS2017 DDoS traffic has strong feature patterns. However, this strength can also become a limitation. A model trained on one attack type may not perform equally well on stealthier or more diverse attacks. Therefore, the result should be presented as a strong initial experiment, not as proof that the model is ready for all network environments.

Future error analysis should inspect the original rows that were misclassified. Important questions include whether the false negatives have feature values close to benign flows, whether the false positives are unusual benign flows, and whether duplicate or near-duplicate records exist. This analysis can support deeper discussion in a thesis chapter or conference paper.

## X. Implementasi Dashboard

Dashboard diimplementasikan menggunakan Flask, Bootstrap, dan Chart.js [14], [15]. It displays total records, normal traffic count, anomaly count, best model accuracy, risk score, risk level, recommended action, model comparison chart, and confusion matrix chart. The prediction page allows users to upload a CSV file and obtain a summary of predicted normal and anomaly records.

This dashboard matters because many beginner administrators understand operational risk more easily through counts, charts, and action recommendations than through raw classification metrics alone. It also creates research artifacts for presentation: screenshots of dashboard, prediction page, confusion matrix, metrics JSON, comparison CSV, and exported prediction result.

## XI. Detail Implementasi

The implementation is separated into preprocessing, training, evaluation, prediction, dashboard, and reporting modules. This separation makes the code easier to inspect. The preprocessing script prepares the data once, while the training script focuses on model comparison and model saving. The evaluation output is stored in JSON and CSV formats to avoid repeated manual copying of metrics into the dashboard.

The prediction module is designed to load the saved model and apply the same feature-handling logic to new CSV files. If a user uploads an incompatible CSV, the system should return a friendly error instead of crashing. This is important for classroom demonstrations because uploaded files may contain different columns, text values, or missing fields.

The dashboard does not use a database in the current version. This design decision keeps the project low-budget and beginner-friendly. The files in the reports folder act as the communication layer between machine learning scripts and the web interface. For future deployment, a database can be added after the research prototype is stable.

## XII. Ancaman terhadap Validitas

The first limitation is dataset scope. The current experiment uses one CICIDS2017 DDoS subset, so it cannot represent all real network conditions. The second limitation is possible benchmark bias because CICIDS2017 has known quality issues discussed in prior literature. The third limitation is that hyperparameter tuning is still simple. The fourth limitation is that the dashboard is a prototype and is not a replacement for enterprise security monitoring.

Despite these limitations, the project is suitable for course and undergraduate research because it demonstrates the complete research cycle: literature analysis, gap identification, novelty formulation, method implementation, experiment, evaluation, dashboard integration, and artifact packaging.

Internal validity is controlled by using fixed random\_state=42 and a clear train-test split. Construct validity is supported by using standard classification metrics. External validity remains limited because one public dataset cannot fully represent campus traffic. Reproducibility validity is supported through README files, scripts, exported metrics, and documented folder structure.

## XIII. Paket Reproducibility

Paket proyek disusun ke dalam lima belas folder utama sesuai struktur artefak UAS AI. The 01\_Paper folder contains primary papers and a Mendeley/IEEE reference guide. The 02\_Literature\_Mapping and 03\_Gap\_Analysis folders contain the literature mapping, comparison matrix, research gap, novelty statement, and research method. The 04\_Dataset folder separates raw and processed data information. The 05\_Source\_Code folder stores scripts and notebooks for preprocessing, training, evaluation, prediction, and dashboard execution.

The 06\_Model folder stores the trained model artifact, while 07\_Hasil\_Eksperimen stores metrics, comparison files, prediction results, and evaluation summaries. The 08\_Visualisasi folder stores charts and diagrams used for the article and presentation. The 09\_Draft\_IEEE folder stores the article draft in DOCX and PDF format, and 10\_Presentasi stores the defense deck and narration. This structure helps another student or examiner reproduce the project from dataset to final report.

Reproducibility also depends on environment documentation. The required commands are pip install -r requirements.txt, python app/preprocessing.py, python app/train.py, and python app/main.py. The random seed is fixed at 42, the train-test split is 80:20, and exported results are stored in machine-readable JSON and CSV formats.

The Google Drive structure is designed so each evidence type can be audited independently. Paper PDFs are separated from mapping documents. Dataset files are separated from source code. Model files are separated from experiment results. Visualizations are separated from presentation materials. This structure reduces the chance that examiners miss an important artifact during assessment.

## XIV. Skenario Deployment Praktis

For a classroom or final project demonstration, NetGuard AI can run locally on a laptop. The Flask dashboard reads result files generated by the training and prediction scripts. This mode is enough for

research defense because the goal is to demonstrate the AI workflow, not to operate a live enterprise monitoring system.

For online demonstration, the application can be deployed on a low-cost VPS, Render, Railway, or similar platform. If the student already owns a domain and uses Cloudflare, the domain can be connected to the deployed server through DNS records. A VPS is more appropriate than a personal laptop because it remains online continuously and does not require the student's device to stay powered on.

For future operational use, the system should be connected to a flow exporter or network sensor that can generate CICIDS-like features. Authentication, upload validation, logging, rate limiting, and secure model storage must be added before exposing the service to public users.

A practical campus deployment can begin with offline batch prediction. Network flow CSV files are exported periodically, uploaded through the dashboard, and analyzed by the trained model. This approach is safer for a student project because it avoids direct packet capture on production networks. Real-time deployment should only be attempted after approval from the institution and after security hardening is completed.

#### XV. Kontribusi Akademik dan Edukatif

The academic contribution of this work is a complete and reproducible student-level research pipeline. The project does not stop at literature review. It includes an implemented machine learning method, baseline comparison, evaluation metrics, visual outputs, and a dashboard prototype. This aligns with the requirement that the assignment must demonstrate implementation and experiment results.

The educational contribution is also important. Students with a networking background can see how raw traffic data becomes flow features, how labels are converted into classes, how machine learning models are trained, and how metrics are interpreted. The dashboard makes the result easier to explain to non-specialist audiences because it connects technical prediction output with operational action.

The project also supports thesis preparation. The current output can be converted into a proposal by expanding the literature review, adding a stronger experimental design, and defining a local validation scenario. It can also be converted into a conference paper by adding cross-validation, cross-dataset testing, and deeper error analysis.

#### XVI. Pengembangan Selanjutnya

Penelitian selanjutnya perlu memperluas dataset melampaui satu subset DDoS. Multi-class classification can include Botnet, Brute Force, PortScan, Web Attack, and Infiltration categories. Cross-dataset evaluation can use UNSW-NB15 or CICIDS2018 to test generalization. Hyperparameter tuning can compare Grid Search, Random Search, and Bayesian Optimization.

Another direction is model interpretability. Feature importance, SHAP values, and error analysis can help explain why a record is classified as anomaly. For deployment, the dashboard can be connected to a lightweight database, authentication system, and scheduled monitoring pipeline. These improvements would move

the project from a research prototype toward a campus monitoring tool.

The web interface can also be developed into a modern full-stack platform. A future version may separate the machine learning API from the frontend, use a database to store experiment history, and implement user roles for administrators and lecturers. However, the current project intentionally avoids this complexity so the core AI research remains understandable.

#### XVII. Kesimpulan

NetGuard AI successfully implements a reproducible machine learning pipeline for network anomaly detection using CICIDS2017. Among Logistic Regression, Decision Tree, and Random Forest, the selected best model is Decision Tree with F1-score 0.9992. The system also provides a lightweight dashboard and risk classification mechanism that translates model output into practical monitoring information.

Future work should validate the system on more attack categories, add cross-validation, perform systematic hyperparameter tuning, test cross-dataset generalization, and evaluate deployment on a small campus network environment under ethical data collection procedures.

Artefak yang paling penting untuk validasi adalah dataset terproses, model.pkl, metrics.json, model\_comparison.csv, confusion matrix, prediction\_result.csv, dashboard screenshot, dan draft artikel. Jika semua artefak tersebut tersedia, maka alur penelitian dapat dilacak mulai dari data, metode, eksperimen, hasil, sampai penyajian. Hal ini mendukung prinsip reproducibility dalam penelitian berbasis AI.

Kesiapan artefak menjadi bagian penting karena tugas ini menuntut penelitian yang dapat direproduksi. Oleh karena itu, folder proyek dipisahkan menjadi paper, literature mapping, gap analysis, dataset, source code, model, hasil eksperimen, visualisasi, draft IEEE, presentasi, Turnitin, deployment, GitHub, dokumentasi, dan bukti submit. Pemisahan ini memudahkan pemeriksaan serta mengurangi risiko file penting tercampur.

#### XIX. Kesiapan Artefak Penelitian

Implementasi berbasis dashboard juga membantu proses demonstrasi. Penguji dapat melihat total record, jumlah trafik normal, jumlah trafik anomali, akurasi model terbaik, risk score, risk level, dan confusion matrix tanpa harus membaca file log mentah. Dengan demikian, dashboard berfungsi sebagai jembatan antara hasil eksperimen machine learning dan kebutuhan pemahaman operasional di bidang jaringan.

Pada lingkungan pendidikan, NetGuard AI dapat digunakan sebagai media pembelajaran integratif antara jaringan komputer, keamanan siber, dan artificial intelligence. Mahasiswa tidak hanya menjalankan model, tetapi juga melihat hubungan antara dataset, fitur trafik, hasil prediksi, metrik evaluasi, dan rekomendasi tindakan. Pendekatan ini membuat pembelajaran lebih konkret karena setiap artefak eksperimen dapat diperiksa ulang oleh dosen maupun mahasiswa lain.

#### XVIII. Implikasi Implementasi pada Lingkungan Pendidikan

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